

$12/8$	$4/8$	$16/8$
$4/8$	1	$4/8$

Fiche

$$1. a) \begin{bmatrix} 1 & -2 & 1 & 2 \\ 1 & 5 & -1 & 1 \\ 1 & 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & -7 & 2 & 1 \\ 0 & -3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & -7 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & -7 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

$$n[A] = v[A|B] = m^0 \text{ de } \text{imaginaris}, \text{ sisteme } \text{posibil } \text{determinate}$$

6)	$\begin{bmatrix} -1 & 1 & 1 \\ 9 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$	$\Rightarrow$	$\begin{bmatrix} -1 & 1 & 1 \\ 0 & 5 & 2 \\ 0 & 2 & 2 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}$	$\begin{bmatrix} -1 & 1 & 1 \\ 0 & 5 & 2 \\ 0 & 0 & -3 \end{bmatrix}$	$\begin{bmatrix} 1 & 5 & 5 \\ 5 & 5 & 2 \\ 5 & 5 & 5 \end{bmatrix}$
----	---	---	---------------	--	---	---	---

$\widehat{[A]} = n[A|B] = m^o$ de incognitas, sistemas possívelmente

c_j	1	1	1	1	1	1	1	1
	1	1	1	1	1	1	1	1
	2	-1	3	2	0	0	-3	1
	4	1	1	4	0	0	0	4

$n[A] = n[A|B] = m$ de imaginația, sistemă posibil de înțeles

$$d) \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 2 & -1 & 1 & -1 & 0 \\ 5 & -1 & 3 & 0 & 0 \\ -1 & 5 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & -3 & 0 \\ 0 & -6 & -2 & -5 & 0 \\ 0 & 6 & 2 & 3 & 0 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & -3 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & -3 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$r[A] = r[A|B] < m^\circ$ de incognitas, sistema possível indeterminado

$$e) \begin{cases} 2x + 3y = y + 3x \\ x - 3z = 2y + 1 \\ 3y + z = 2 - 2x \end{cases} \Rightarrow \begin{cases} -x + 2y = 0 \\ x + 2y - 3z = 1 \\ 2x + 3y + z = 2 \end{cases}$$

$$\begin{bmatrix} -1 & 2 & 0 & 0 \\ 1 & -1 & -3 & 1 \\ 2 & 3 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 0 & 0 \\ 0 & 7 & 1 & 1 \\ 0 & 0 & -3 & 1 \end{bmatrix}$$

$r[A] = r[A|B] = m^\circ$ de incognitas, sistema possível determinado

$$f) \begin{bmatrix} 3 & 1 & 1 & 1 & 0 \\ 2 & 0 & -1 & 1 & 1 \\ 1 & 1 & -2 & -1 & -1 \\ 3 & -1 & 5 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 1 & 1 & 1 & 0 \\ 0 & 2 & 5 & -4 & -3 \\ 0 & -2 & 7 & 4 & 3 \\ 0 & 2 & -4 & 1 & -2 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} 3 & 1 & 1 & 1 & 0 \\ 0 & 2 & 5 & -4 & -3 \\ 0 & 0 & 12 & 0 & 0 \\ 0 & 0 & 9 & -5 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 1 & 1 & 1 & 0 \\ 0 & 2 & 5 & -4 & -3 \\ 0 & 0 & 12 & 0 & 0 \\ 0 & 0 & 0 & 60 & -70 \end{bmatrix}$$

$r[A] = r[A|B] = m^\circ$ incognitas, sistema possível determinado

$$g) \begin{bmatrix} 1 & -3 & 1 & 2 \\ 2 & -4 & 3 & 7 \\ -3 & 1 & 2 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 1 & 2 \\ 0 & 2 & 1 & 5 \\ 0 & -8 & 5 & 13 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 1 & 2 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 9 & 53 \end{bmatrix}$$

$r[A] = r[A|B] = m^\circ$ de incógnitas, sistema possível determinado

$$h) \begin{bmatrix} 1 & 1 & -1 & 3 \\ 2 & -1 & 3 & 13 \\ 3 & -3 & 7 & 23 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & -3 & 5 & 7 \\ 0 & -6 & 10 & 14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & -3 & 5 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$r[A] = r[A|B] < m^\circ$ de incógnitas, sistema possível indeterminado

$$i) \begin{bmatrix} -2 & 1 & 3 & 10 \\ 10 & -5 & -15 & 30 \\ 1 & 1 & -3 & 25 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 1 & 3 & 10 \\ 0 & 0 & 0 & 80 \\ 0 & 3 & -3 & 60 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} -2 & 1 & 3 & 10 \\ 0 & 3 & -3 & 60 \\ 0 & 0 & 0 & 80 \end{bmatrix}$$

$r[A] < r[A|B] = m^\circ$ de incógnitas, sistema impossível

$$j) \begin{bmatrix} 2 & -5 & 4 & -3 \\ 1 & -2 & 1 & 5 \\ 1 & -3 & 3 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -5 & 4 & -3 \\ 0 & -1 & 2 & -13 \\ 0 & 1 & -2 & 13 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -5 & 4 & -3 \\ 0 & -1 & 2 & -13 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$r[A] = r[A|B] < m^\circ$ de incógnitas, sistema possível indeterminado

$$2. A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 2 \\ 1 & -3 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 5 \\ 0 & 5 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r[A] = 2$$

$$B = \begin{bmatrix} 6 & 3 & 2 \\ 0 & 1 & 2 \\ 3 & 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 3 & 2 \\ 0 & 1 & 2 \\ 0 & -5 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 3 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 8 \end{bmatrix} \quad r[B] = 3$$

$$C = \begin{bmatrix} 1 & 1 & 1 & 2 & 0 & -1 \\ 1 & 0 & 2 & -1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 2 & 0 & -1 \\ 0 & 1 & -1 & 3 & -1 & -2 \\ 0 & 0 & 1 & 2 & -1 & 0 \end{bmatrix}$$

$$r[C] = 3$$

$$D = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 0 & -5 & -8 \\ 0 & -3 & 0 \\ 0 & 1 & 2 \\ 0 & -3 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 0 & -5 & -8 \\ 0 & 0 & -8 \\ 0 & 0 & 2 \\ 0 & 0 & \frac{20}{3} \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 0 & -5 & -8 \\ 0 & 0 & -8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r[D] = 3$$

$$3. A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & a-1 & a \\ a & 0 & a \\ a+1 & 0 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & a-1 & a \\ 0 & 0 & -a \\ 0 & 0 & 4+2(a+1) \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & a-1 & a \\ 0 & 0 & -a \\ 0 & 0 & 0 \end{bmatrix}$$

$$Se \ a = 1, \ r[A] = 2$$

$$Se \ a = 0, \ r[A] = 2$$

$$Se \ a \neq 0 \vee a \neq 1, \ r[A] = 3$$

$$B = \begin{bmatrix} \beta & -a & 0 \\ 0 & \beta & 0 \\ 0 & 0 & a \end{bmatrix}$$

$$Se \ a = 0 \wedge \beta \neq 0 \quad r[B] = 2$$

$$Se \ \beta = 0 \wedge a \neq 0 \quad r[B] = 2$$

$$Se \ a = 0 \wedge \beta = 0 \quad r[B] = 0$$

$$Se \ a \neq 0 \wedge \beta \neq 0 \quad r[B] = 3$$

$$4. a) \begin{bmatrix} 1 & -1 & 1 & -1 \\ 2 & 0 & 1 & 2 \\ 1 & -1 & a & \beta \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 2 & -1 & 4 \\ 0 & 0 & a-1 & \beta+1 \end{bmatrix}$$

$$Se \ a = 1 \wedge \beta = -1$$

$r[A] = r[A|B] < m^\circ$ de incógnitas, sistema possível indeterminado

Se $a = 1 \wedge \beta \neq 1$

$r[A] < r[A|B] = \text{m}^\circ \text{ de incógnitas, sistema impossível}$

Se $a \neq 1 \wedge \beta = 1$

$r[A] = r[A|B] = \text{m}^\circ \text{ de incógnitas, sistema possível determinável}$

Se $a \neq 1 \wedge \beta \neq 1$

$r[A] = r[A|B] = \text{m}^\circ \text{ de incógnitas, sistema possível determinável}$

b)
$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & -3 \\ -1 & 4 & -1 & 3a \\ \beta & 0 & 1 & 3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & -3 \\ 0 & 3 & 0 & 3(a-1) \\ 0 & \beta & 1-\beta & 3(1+\beta) \end{array} \right] \rightarrow$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & -3 \\ 0 & 3 & 0 & 3(a-1) \\ 0 & 0 & 1-\beta & 3+3\beta-3a\beta-\beta \end{array} \right]$$

Se $\beta \neq 1$,

$r[A] = r[A|B] = \text{m}^\circ \text{ de incógnitas, sistema possível determinável}$

Se $\beta = 1$

$r[A] < r[A|B]$, sistema impossível

c)
$$\left[\begin{array}{ccc|c} 1 & -a & 1 & -\beta \\ 1 & -1 & \beta+1 & 1 \\ 1 & -1 & 1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -a & 1 & -\beta \\ 0 & 1+a & -\beta & -(\beta+1) \\ 0 & 1+a & 0 & -(\beta+3) \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -a & 1 & -\beta \\ 0 & 1+a & -\beta & -(\beta+1) \\ 0 & 0 & \beta & -\beta-3+\beta+1 \end{array} \right]$$

Se $a = -1 \wedge \beta = 0$

$r[A] < r[A|B]$, sistema impossível

Se $a = -1 \wedge \beta = -1$

$r[A] < r[A|B]$, sistema impossível

Se $a \neq -1 \wedge B \neq 0$

$r[A] < r[A|B]$, sistema impossível

Se $a \neq -1 \wedge B \neq 0$

$r[A] = r[A|B] = \text{m}^\circ \text{ de incógnitas}$, sistema possível determinado

$$d) \left[\begin{array}{ccc|c} -2 & a & -B & -3 \\ 1 & 0 & B & 3a \\ 2 & 4 & 3B & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} -2 & a & -B & -3 \\ 0 & a & B & 6a-3 \\ 0 & a+4 & 2B & 0 \end{array} \right] \rightarrow$$

$$\rightarrow \left[\begin{array}{ccc|c} -2 & a & -B & -3 \\ 0 & a & B & 6a-3 \\ 0 & 0 & -1 & 3(2a-1)(a+4) \end{array} \right]$$

$\downarrow$
 $-2Ba + B(a+4)$

Se $a \neq 0 \wedge B = 0$

$r[A] < r[A|B]$, sistema impossível

Se $a = 0 \wedge B = 0$

$r[A] < r[A|B]$, sistema impossível

Se $a = 0 \wedge B \neq 0$

$r[A] = r[A|B] < \text{m}^\circ \text{ de incógnitas}$, sistema possível indeterminado

Se $a \neq 0 \wedge B \neq 0$

$r[A] = r[A|B] = \text{m}^\circ \text{ de incógnitas}$, sistema possível determinado

$$5) \left[\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 0 \\ 0 & 2 & -2 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 0 & -6 & 0 & -2 & 1 & 0 \\ 0 & 2 & -2 & 0 & 0 & 1 \end{array} \right] \rightarrow$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 0 & -6 & 0 & -2 & 1 & 0 \\ 0 & 0 & -6 & -2 & 1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 6 & 0 & 0 & -2 & 4 & 0 \\ 0 & -6 & 0 & -2 & 1 & 0 \\ 0 & 0 & -6 & -2 & 1 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/3 & 2/3 & 0 \\ 0 & 1 & 0 & 1/3 & -1/6 & 0 \\ 0 & 0 & 1 & 1/3 & -1/6 & -1/2 \end{array} \right]$$

$$6) \begin{bmatrix} 2 & 1 & -2 & 1 & 0 & 0 \\ -2 & 1 & 2 & 0 & 1 & 0 \\ -2 & -4 & 4 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & -2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & -3 & 2 & 1 & 0 & 1 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} 2 & 1 & -2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 4 & 5 & 3 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 2 & 0 & 7 & 3 & 2 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 4 & 5 & 3 & 2 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} 4 & 0 & 0 & 6 & 2 & 2 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 4 & 5 & 3 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 3/2 & 1/2 & 1/2 \\ 0 & 1 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & 5/4 & 3/4 & 1/2 \end{bmatrix}$$

$$6. a) x = \begin{vmatrix} 6 & -1 \\ 2 & 5 \\ 2 & -1 \\ 4 & 5 \end{vmatrix} = \frac{32}{14} = \frac{16}{7} \quad y = \begin{vmatrix} 6 & 2 \\ 2 & 4 \\ 2 & -1 \\ 4 & 5 \end{vmatrix} = \frac{20}{14} = \frac{10}{7}$$

$$b) x = \begin{vmatrix} -1 & 1 & 1 \\ -5 & -4 & 2 \\ 2 & -2 & 1 \end{vmatrix} = -1 \begin{vmatrix} -4 & 2 \\ -2 & 1 \end{vmatrix} - 1 \begin{vmatrix} -5 & 2 \\ 2 & 1 \end{vmatrix} + 1 \begin{vmatrix} -5 & -4 \\ 2 & -2 \end{vmatrix} = -1(-4-2) - 1(-5-4) + 1(10-8) = 6 + 9 + 2 = 17$$

$$= \frac{17}{9} = 3$$

$$y = \begin{vmatrix} -1 & 1 & 1 \\ -5 & 3 & 2 \\ -2 & 3 & 1 \end{vmatrix} = -1 \begin{vmatrix} 3 & 2 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} -5 & 2 \\ -2 & 1 \end{vmatrix} + \begin{vmatrix} -5 & 3 \\ -2 & 3 \end{vmatrix} = -1(3-6) + (-5-2) + (-15-6) = 3 - 7 - 21 = -25$$

$$z = \begin{vmatrix} -1 & 1 & 1 \\ -5 & 3 & -4 \\ -2 & 3 & -2 \end{vmatrix} = -1 \begin{vmatrix} 3 & -4 \\ 3 & -2 \end{vmatrix} - \begin{vmatrix} -5 & -4 \\ -2 & -2 \end{vmatrix} + \begin{vmatrix} -5 & 3 \\ -2 & 3 \end{vmatrix} = -1(-6-12) - (10-8) + (-15-6) = 18 - 2 - 21 = -5$$

c) $\alpha =$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & -1 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} - \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix} = -6 - (-6) = 0$$

$\beta =$

$$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 3 & -1 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 3 & -1 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = -6 - 6 - 13 = -25$$

$\gamma =$

$$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 3 & 2 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = -6 - 6 - 13 = -35$$